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The Explabox: Model-Agnostic Machine Learning Transparency & Analysis

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Summary

We present the `Explabox`: an open-source toolkit for transparent and responsible machine learning (ML) model development and usage. `Explabox` aids in achieving explainable, fair and robust models by employing a four-step strategy: *explore*, *examine*, *explain* and *expose*. These steps offer model-agnostic analyses that transform complex `ingestibles` (models and data) into interpretable `digestibles`. The toolkit encompasses digestibles for descriptive statistics, performance metrics, model behavior explanations (local and global), and robustness, security, and fairness assessments. Implemented in Python, `Explabox` supports multiple interaction modes and builds on open-source packages. It empowers model developers and testers to operationalize explainability, fairness, auditability, and security. The initial release focuses on text data and models, with plans for expansion. `Explabox`'s code and documentation are available open-source at https://explabox.readthedocs.io.

Statement of need

It is crucial that Machine Learning (ML) development and usage is done in a responsible and transparent manner. High-stakes organizational decisions may significantly impact individuals and society, with potential severe consequences stemming from biases or model errors. This is exemplified by the EU AI Act's regulatory framework, requiring high-risk systems to be properly tested, documented and assessed on their conformity before being applied in practice (Edwards, 2022). Yet, operationalizing transparency (explainable ML) and testing model behavior (fairness, robustness, and security auditing) remains a difficult and laborious task, given the myriad of techniques and their associated learning curves. In response, we have devised a comprehensive four-step analysis strategy—*explore*, *examine*, *explain*, and *expose*—ensuring holistic model transparency and testing. The open-source `Explabox` offers these analyses through well-documented, reproducible steps. Data scientists can now access a unified, model-agnostic approach developed and used in a high-stakes context—the Netherlands National Police—that is applicable to any text classifier or regressor.

We have developed the `Explabox` in an organizational environment where models and data are analyzed repeatedly, and where internal and external stakeholders have varying explanatory needs and preferred formats. Several related tools have been made available, such as `AIX360` (Arya et al., 2019), `alibi explain` (Klaise et al., 2021), `dalex` (Baniecki et al., 2021), `CheckList` (Ribeiro et al., 2020), and `AIF360` (Bellamy et al., 2018). However, these tools exhibit shortcomings such as incompatibility with recent Python versions (3.8–3.12), restricted software functionality primarily focused on testing or explainability, an absence of reproducible outcomes, or the lack of means to provide the flexibility regarding how results can be communicated to address stakeholder needs. To fill this gap, we propose the `Explabox`. The `Explabox` is an open-source

Python toolkit that supports organizations with responsible ML development with minimal disruption to practitioners' workflows through a four-step analysis strategy easily embedded with existing datasets and models, and providing a central node for connecting with state-of-the-art research and sharing best practices.

Explore, Examine, Explain & Expose your ML models

The `Explabox` transforms opaque *ingestibles* into transparent *digestibles* through four types of *analyses*. The digestibles provide insights into model behavior and data, enhancing model explainability and assisting in auditing the fairness, robustness, and security of ML systems.

Ingestibles

Ingestibles serve as a unified interface for importing models and data. The layers (Figure 1) abstract away from how the model and data are accessed, and allow for optimized processing. The `Explabox` encapsulates the model and data with `instancelib` (Bron, 2023) to ensure fast processing. The model can be any Python `Callable` containing a regression or (binary and multi-class) classification model. Models developed with `scikit-learn` (Pedregosa et al., 2011) or with inferencing through `onnx` (e.g., PyTorch and TensorFlow/Keras) can be imported directly with further optimizations and automatic extraction of how inputs/outputs are to be interpreted. Data can be automatically downloaded, extracted and loaded. Data can be provided as `NumPy` arrays, `Pandas` DataFrames, `huggingface` datasets, raw files (e.g., HDF5, CSV or TSV), or as (compressed) folders containing raw files. The data can be subdivided into named splits (e.g., train-test-validation), and instance vectors and tokens can be precomputed (and optionally saved on disk) to provide fast inferencing.

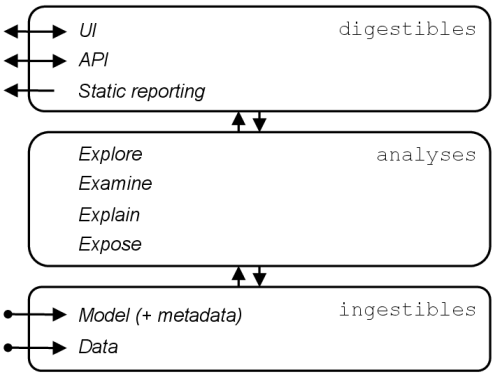

Figure 1: Logical separation of the `Explabox` into layers with interfaces.

Analyses

The `Explabox` turns these *ingestibles* into *digestibles*: pieces of information that increase the transparency of the ingestibles. Turning ingestibles into digestibles is done through four types of analyses: **explore**, **examine**, **explain** and **expose**.

Explore allows slicing, dicing and sorting data, and provides descriptive statistics, grouped by named split. Relevant statistics include data set sizes and label distributions, and modality-relevant information, such as string lengths and tokenized lengths for textual data.

Examine shows performance metrics of the ML model on the data, summarized in a table or shown graphically. Metrics are accompanied by references on how they are computed and how they should be interpreted. For further analysis, **examine** also supports to drill-down into which instances were predicted correctly and incorrectly.

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Explain uses model-agnostic techniques (Ribeiro et al., 2016a) to explain model behavior (*global*) and individual predictions (*local*). It summarizes model-labelled data, through prototypes (`K-Medoids`) or prototypes and criticisms (`MMDCritic` (Kim et al., 2016)), and token distributions (`TokenFrequency`) and informativeness (`TokenInformation`) for the text modality. Local explanations are given by popular techniques for feature attribution scores (`LIME` (Ribeiro et al., 2016b), `KernelSHAP` (Lundberg & Lee, 2017)), relevant feature subsets (`Anchors` (Ribeiro et al., 2018)), local rule-based models (`LORE` (Guidotti et al., 2018)), and counterfactual/contrastive explanations (`FoilTrees` (Waa et al., 2018)). These methods are constructed from generic components that split the relevant steps in global and local explanation generation. This allows customizability and for scientific advancements to quickly end up in operational processes. For example, they may combine the data sampling from `KernelSHAP` and summarize these with a surrogate rule-based model provided by the `imodels` package (Singh et al., 2021). However, to ease users into adoption we provide example configurations, such as `LIME` with default hyperparameters. **Explain** is provided by subpackage `text_explainability` (Robeer, 2021a), which doubles as a standalone tool.

Expose gathers sensitivity insights through local/global testing regimes. These insights can be used to, through relevant attributes, assess the *robustness* (e.g., the effect of typos on model performance), *security* (e.g., if inputs containing certain characters crash the model) and *fairness* (e.g., subgroup performance for protected attributes such as country of origin, gender, race or socioeconomic status) of the model. Relevant attributes can either be observed in the current data or generated from user-provided templates (Ribeiro et al., 2020). These attributes are then either summarized in performance metrics, compared to expected behavior (Ribeiro et al., 2020), or assessed with fairness metrics for classification (Mehrabi et al., 2021) and regression (Agarwal et al., 2019). Like **explain**, **expose** is also made from generic components, which allows users to customize data generation and tests. **Expose** is provided by the `text_sensitivity` subpackage (Robeer, 2021b), which also doubles as a standalone tool.

Digestibles

To serve diverse stakeholders' needs—such as auditors, applicants, end-users or clients (Tomsett et al., 2018)—in consuming model and data insights, the digestibles are accessible through different channels like an interactive user interface (UI) for Jupyter Notebook or webpages (Figure 2), an API for integration with other tooling, and static reporting.

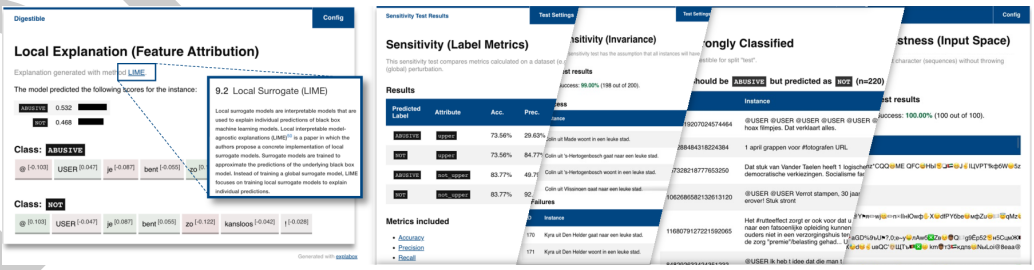

Figure 2: UI elements of the Jupyter Notebook interface for interactive explainability and analyses.

Open-source implementation

`Explabox` is a Python library with full cross-platform support for versions 3.8–3.12 (see Figure 3 for example analyses). Distributed under the GNU LGPL-3.0 license, it offers a flexible inferencing and data-handling API through `instancelib` (Bron, 2023). It also benefits from other open-source communities, such as `scikit-learn` (Pedregosa et al., 2011) for surrogate models, clustering and dimensionality reduction, `imodels` (Singh et al., 2021) for rule-based interpretable models, `Faker` (Faraglia & Other Contributors, 2021) for multi-language data generation, and `plotly` (Plotly Technologies Inc., 2015) for interactive and static graphics.

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The Explabox documentation provides installation guides, a tutorial to get started, a comprehensive example use-case using Jupyter Notebook, a full overview of all Python classes and functions, and guides on how to contribute. Documentation generation, testing, code quality assurance, and versioning are automated with a GitHub CI/CD pipeline.

```python
# Wrap your model & data
from explabox import Explabox
box = Explabox(data=data, model=model)

# Explore data distributions
box.explore()

# Examine model performance on the data
box.examine.wrongly_classified()

# Explain local/global model behavior
box.explain.explain_prediction('Explain this!')

# Expose where the model is sensitive to
box.expose.compare_metrics(perturbation='upper')
```

Figure 3: Python snippet with example analyses, to facilitate model-agnostic responsible AI development with explainability and versatile analysis tools.

Conclusion and Future Work

The Explabox enables organizations to responsibly develop and apply AI applications. The model-agnostic toolkit provides interactive analyses that model developers and testers can use to operationalize, report and discuss model explainability, fairness, auditability, and security. It provides flexibility to handle various ML use-cases for classification and regression, while also being a central node for standardization and connecting research with practice. The first full release, focusing on text data and models, is available open-source and fully documented at https://explabox.readthedocs.io.

Further development has already started, where we focus on extending the Explabox for (1) models and data with the tabular, audio and image modality, and multi-modal combinations; (2) improved integration of data and model provenance for auditability; (3) comparisons of multiple datasets and multiple models, and; (4) research and development for a new interface and improved stakeholder data visualization.

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References

Agarwal, A., Dudik, M., & Wu, Z. S. (2019). Fair Regression: Quantitative Definitions and Reduction-Based Algorithms. In K. Chaudhuri & R. Salakhutdinov (Eds.), *Proceedings of the 36th international conference on machine learning* (Vol. 97, pp. 120–129). PMLR.

Arya, V., Bellamy, R. K. E., Chen, P.-Y., Dhurandhar, A., Hind, M., Hoffman, S. C., Houde, S., Liao, Q. V., Luss, R., Mojsilović, A., Mourad, S., Pedemonte, P., Raghavendra, R., Richards, J., Sattigeri, P., Shanmugam, K., Singh, M., Varshney, K. R., Wei, D., & Zhang, Y. (2019). One Explanation Does Not Fit All: A Toolkit and Taxonomy of AI Explainability Techniques. *arXiv Preprint*. https://doi.org/arXiv.1909.03012

Baniecki, H., Kretowicz, W., Piatyszek, P., Wisniewski, J., & Biecek, P. (2021). dalex: Responsible Machine Learning with Interactive Explainability and Fairness in Python. *Journal of Machine Learning Research*, *22*(214), 1–7. http://jmlr.org/papers/v22/20-1473.html

Bellamy, R. K. E., Dey, K., Hind, M., Hoffman, S. C., Houde, S., Kannan, K., Lohia, P., Martino, J., Mehta, S., Mojsilovic, A., Nagar, S., Ramamurthy, K. N., Richards, J., Saha, D., Sattigeri, P., Singh, M., Varshney, K. R., & Zhang, Y. (2018). AI Fairness 360: An Extensible Toolkit for Detecting, Understanding, and Mitigating Unwanted Algorithmic Bias. *arXiv Preprint*. https://doi.org/arXiv.1810.01943

Bron, M. P. (2023). *Python Package instancelib* (Version 0.5.0). Zenodo. https://doi.org/10.5281/zenodo.8308017

Edwards, L. (2022). The EU AI Act: A summary of its significance and scope. *Ada Lovelace Institute, Expert Explainer Report*. https://www.adalovelaceinstitute.org/wp-content/uploads/2022/04/Expert-explainer-The-EU-AI-Act-11-April-2022.pdf

Faraglia, D., & Other Contributors. (2021). *Python package Faker*. https://github.com/joke2k/faker

Guidotti, R., Monreale, A., Ruggieri, S., Pedreschi, D., Turini, F., & Giannotti, F. (2018). Local Rule-Based Explanations of Black Box Decision Systems. *arXiv Preprint*. https://doi.org/arXiv.1805.10820

Kim, B., Khanna, R., & Koyejo, O. (2016). Examples are not Enough, Learn to Criticize! Criticism for Interpretability. *29th Conference on Neural Information Processing Systems (NIPS 2016)*.

Klaise, J., Looveren, A. V., Vacanti, G., & Coca, A. (2021). Alibi Explain: Algorithms for Explaining Machine Learning Models. *Journal of Machine Learning Research*, *22*(181), 1–7. http://jmlr.org/papers/v22/21-0017.html

Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 4765–4774.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys (CSUR)*, *54*(6), 1–35.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Plotly Technologies Inc. (2015). *Collaborative data science*. Plotly Technologies Inc. https://plot.ly

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016a). Model-Agnostic Interpretability of Machine Learning. *2016 ICML Workshop on Human Interpretability in Machine Learning (WHI 2016)*, 91–95. https://doi.org/10.1145/2858036.2858529

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016b). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *22nd ACM SIGKDD International Conference on Knowledge Discovery in Data Mining (KDD'16), Proceedings*, 1135–1144. ISBN: 9781450321389

Ribeiro, M. T., Singh, S., & Guestrin, C. (2018). Anchors: High-Precision Model-Agnostic

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Explanations. *AAAI Conference on Artificial Intelligence, Proceedings*.

Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond Accuracy: Behavioral Testing of NLP models with CheckList. *Association for Computational Linguistics (ACL)*.

Robeer, M. (2021a). *Python package text_explainability*. https://doi.org/10.5281/zenodo.14192126

Robeer, M. (2021b). *Python package text_sensitivity*. https://doi.org/10.5281/zenodo.14192940

Singh, C., Nasseri, K., Tan, Y. S., Tang, T., & Yu, B. (2021). imodels: A python package for fitting interpretable models. *Journal of Open Source Software*, *6*(61), 3192. https://doi.org/10.21105/joss.03192

Tomsett, R., Braines, D., Harborne, D., Preece, A., & Chakraborty, S. (2018). Interpretable to Whom? A Role-based Model for Analyzing Interpretable Machine Learning Systems. *2018 ICML Workshop on Human Interpretability in Machine Learning (WHI 2018)*.

Waa, J. van der, Robeer, M., Diggelen, J. van, Neerincx, M., & Brinkhuis, M. (2018). Contrastive Explanations with Local Foil Trees. *2018 Workshop on Human Interpretability in Machine Learning (WHI 2018)*.

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